

Q18) Write a C program for DES the first 24 bits of each subkey come from the same subset of 28 bits of the initial key and that the second 24 bits of each subkey come from a disjoint subset of 28 bits of the initial key.

Aim: To write a C program to demonstrate the **DES key generation process**, showing that the first 24 bits of each subkey are selected from the left 28-bit half (C) and the remaining 24 bits are selected from the right 28-bit half (D) of the original 56-bit key.

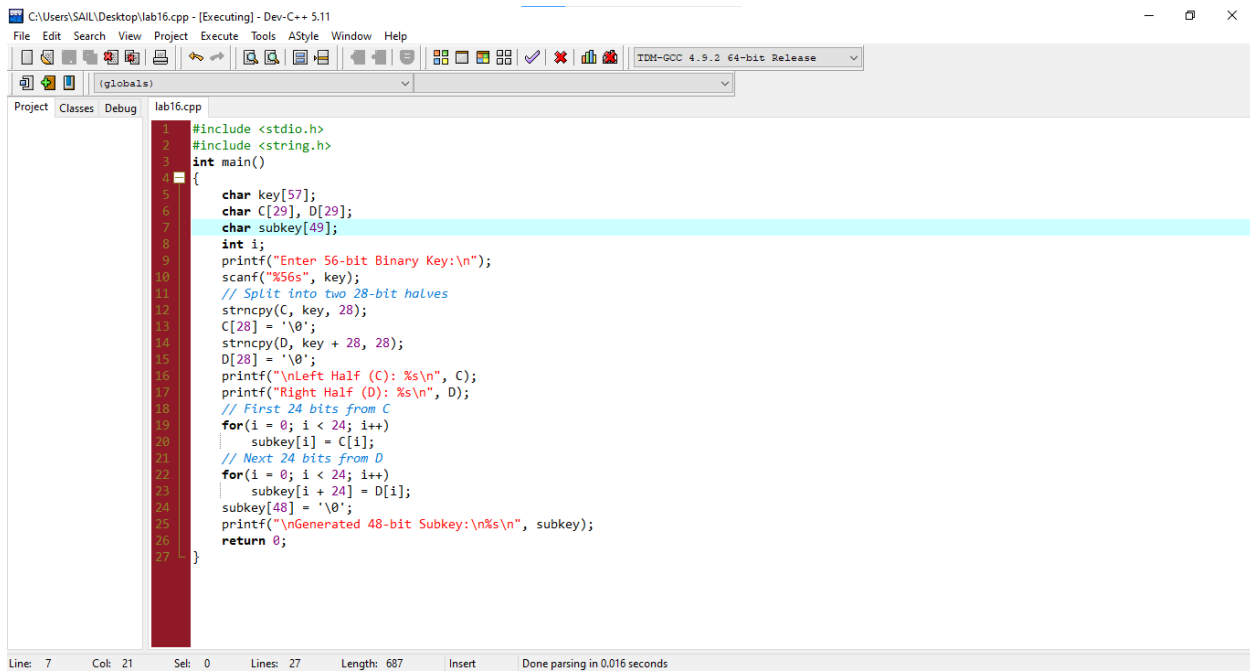
Algorithm:

1. Start the program.
2. Enter a 56-bit binary key.
3. Split the key into two equal halves: **C (28 bits)** and **D (28 bits)**.
4. Display both halves separately.
5. Select the first 24 bits from the left half (C).
6. Select the first 24 bits from the right half (D).
7. Combine these two 24-bit parts to form a 48-bit DES subkey.
8. Display the generated subkey.
9. Repeat the process for additional rounds if required.
10. Stop the program.

Result:

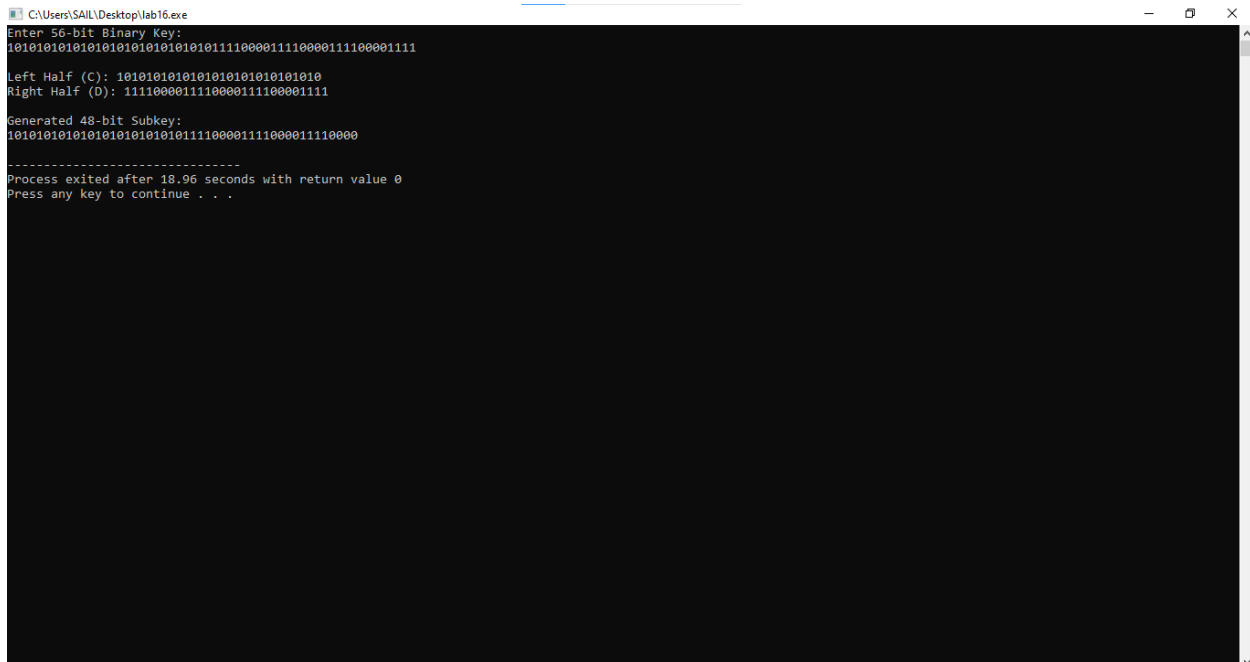
Thus, the C program to demonstrate the **DES key generation process** was successfully executed. It was observed that the **first 24 bits** of the generated subkey were obtained from the **left 28-bit half (C)**, while the **remaining 24 bits** were obtained from the **right 28-bit half (D)**, illustrating the DES subkey generation concept.

Code:



```
1 #include <stdio.h>
2 #include <string.h>
3 int main()
4 {
5     char key[57];
6     char C[29], D[29];
7     char subkey[49];
8     int i;
9     printf("Enter 56-bit Binary Key:\n");
10    scanf("%56s", key);
11    // Split into two 28-bit halves
12    strncpy(C, key, 28);
13    C[28] = '\0';
14    strncpy(D, key + 28, 28);
15    D[28] = '\0';
16    printf("\nLeft Half (C): %s\n", C);
17    printf("Right Half (D): %s\n", D);
18    // First 24 bits from C
19    for(i = 0; i < 24; i++)
20        subkey[i] = C[i];
21    // Next 24 bits from D
22    for(i = 0; i < 24; i++)
23        subkey[i + 24] = D[i];
24    subkey[48] = '\0';
25    printf("\nGenerated 48-bit Subkey:\n%s\n", subkey);
26    return 0;
27 }
```

Output:



```
C:\Users\SAIL\Desktop\lab16.exe
Enter 56-bit Binary Key:
101010101010101010101010111000011110000111100001111

Left Half (C): 10101010101010101010101010101010
Right Half (D): 1111000011110000111100001111

Generated 48-bit Subkey:
101010101010101010101010111100001111000011110000

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Process exited after 18.96 seconds with return value 0
Press any key to continue . . .
```